

Algorithms

$$L = \{ w \# w \mid w \in \{0,1\}^* \}$$

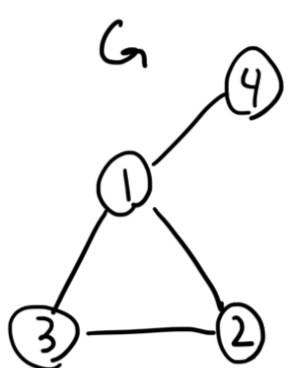
$M_1 =$ "On input w :

1. zig-zag across the tape to corresponding position on either side of the $\#$ to check if they are the same symbol.
If not, or no $\#$ exists, **reject**.
Cross off symbols to check if they've been visited.
2. When all symbols to the left of $\#$ have been crossed off, check for any remaining symbols to the right of $\#$. If any symbols remain, **reject**, otherwise **accept**.

"Snapshot history"

↓
011000#011000 ⊔ ...
 ↓
x11000#011000 ⊔
 ↓
x11000#x11000 ⊔
 ↓
xx1000#x11000 ⊔
 ⋮
xxxxxx#xxxxxx ⊔
 ↓
 accept

Any algorithm can be expressed via a TM
We can begin to ignore low-level implementation details.



"encoding"

$\langle G \rangle =$

$(1,2,3,4) \underbrace{((1,2)(2,3)(3,1)(1,4))}_{\text{edges}}$
 vertices

Is G connected?

$M =$ "On input $\langle G \rangle$, encoding of graph G :

1. Select first node & mark it.
2. Repeat 3 until no new nodes are marked:
3. For each node in G , mark it if it is adjacent to a marked node.
4. Scan all nodes. If any are not marked, **reject**. Otherwise **accept**.

(M decides if a graph is connected.)

" M decides the language of connected graphs."